

APR - 18/BE/Insem. - 26

B.E. (Mechanical)

FINITE ELEMENT ANALYSIS

(2012 Course) (Elective - IV) (Semester - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

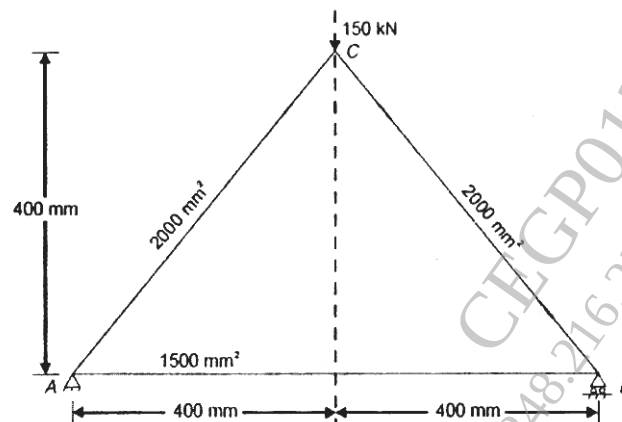
- 1) Draw suitable neat diagrams, wherever necessary.
- 2) Figures to the right indicate full marks.
- 3) Use of electronic pocket calculator is allowed.
- 4) Assume suitable data, if required.

- Q1)** a) Write down step by step procedure for FEA and explain importance of convergence. [6]
 b) Give two examples of plane stress problem and its significance in FEA simulations. [4]

OR

- Q2)** a) Explain difference between Finite Element and Finite Difference Method. [6]
 b) Explain terms in Weighted Residual Method and significance of weight function. [4]

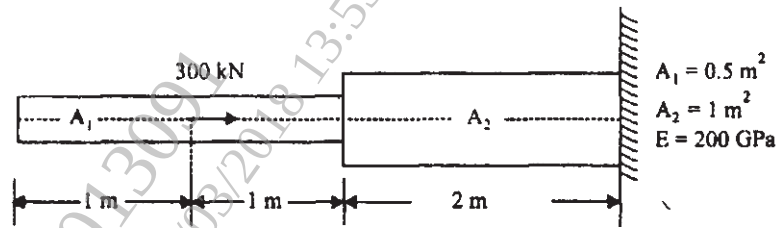
- Q3)** Determine the nodal displacements and the stress in each member. Find the support reactions also. Take modulus of elasticity as 200 GPa. Taking node 1 as the origin, the coordinates of various nodes are 1 (0, 0), 2 (800, 0), 3 (400, 400). [10]



OR

P.T.O.

- Q4) a)** Determine the nodal displacements and element stresses by finite element formulation for the following figure. Use $P = 300 \text{ kN}$; $A_1 = 0.5 \text{ m}^2$; $A_2 = 1 \text{ m}^2$; $E = 200 \text{ GPa}$. [6]



- b) Explain the elimination method and penalty method of imposing boundary conditions. Comment on the two methods. [4]

- Q5) a)** Write a Note on CST element and write a equation of stiffness matrix. [4]

- b) Write a note on Pascal's Triangle for identify interpolation function for 2nd order element. [6]

OR

- Q6) a)** What is meant by Banded and Skyline Matrix methods and how these are used for reduction in memory required to simulation in FEA? [6]

- b) Explain significance of Axis-symmetric element for conversion of 3D to 2D problem and its practical application. [4]

